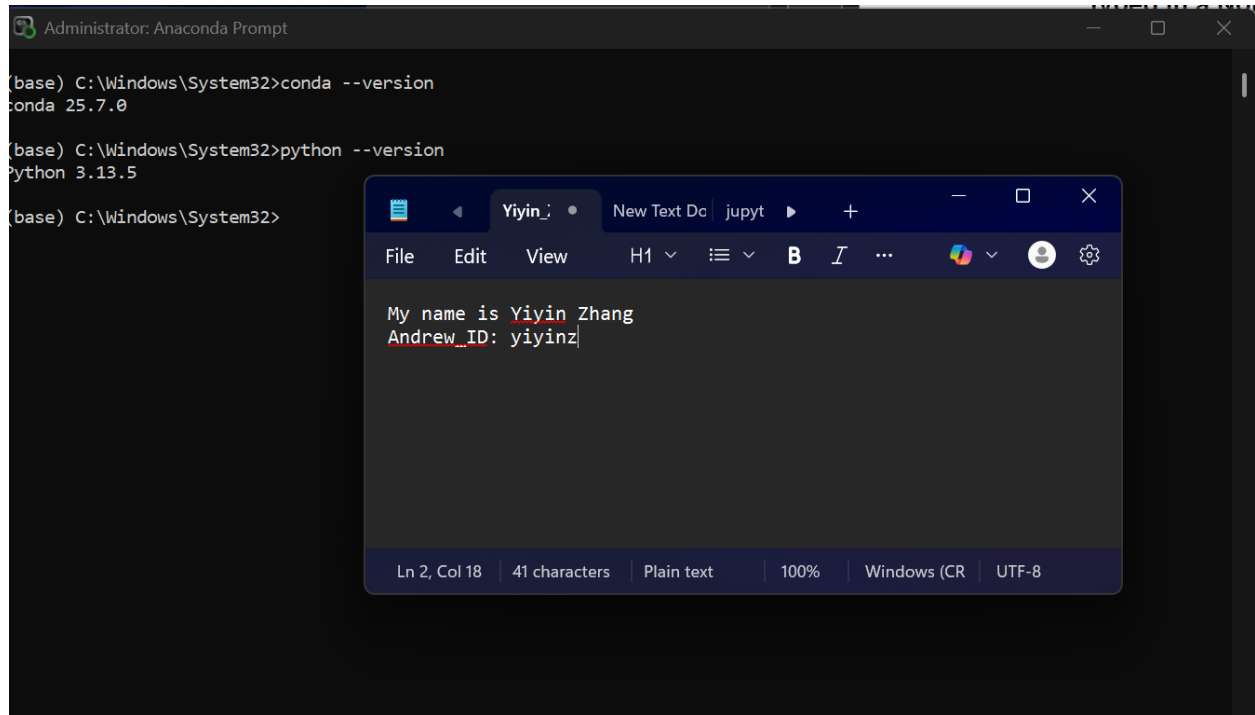


Part-I

Anaconda Python



The image shows a screenshot of an Anaconda Prompt window and a Jupyter Notebook editor. The Anaconda Prompt window is titled "Administrator: Anaconda Prompt" and shows the following commands and output:

```
(base) C:\Windows\System32>conda --version
conda 25.7.0

(base) C:\Windows\System32>python --version
Python 3.13.5

(base) C:\Windows\System32>
```

Overlaid on the prompt is a Jupyter Notebook editor window titled "Yiyin_". The notebook contains the following text:

```
My name is Yiyin Zhang
Andrew_ID: yiyinz
```

The Jupyter Notebook editor window has a menu bar with "File", "Edit", and "View". The status bar at the bottom indicates "Ln 2, Col 18", "41 characters", "Plain text", "100%", "Windows (CR)", and "UTF-8".

Jupyter

The screenshot displays the Jupyter Notebook interface within the Anaconda Toolbox. The main window shows a Jupyter Notebook titled "Untitled" with a code cell containing the following Python code:

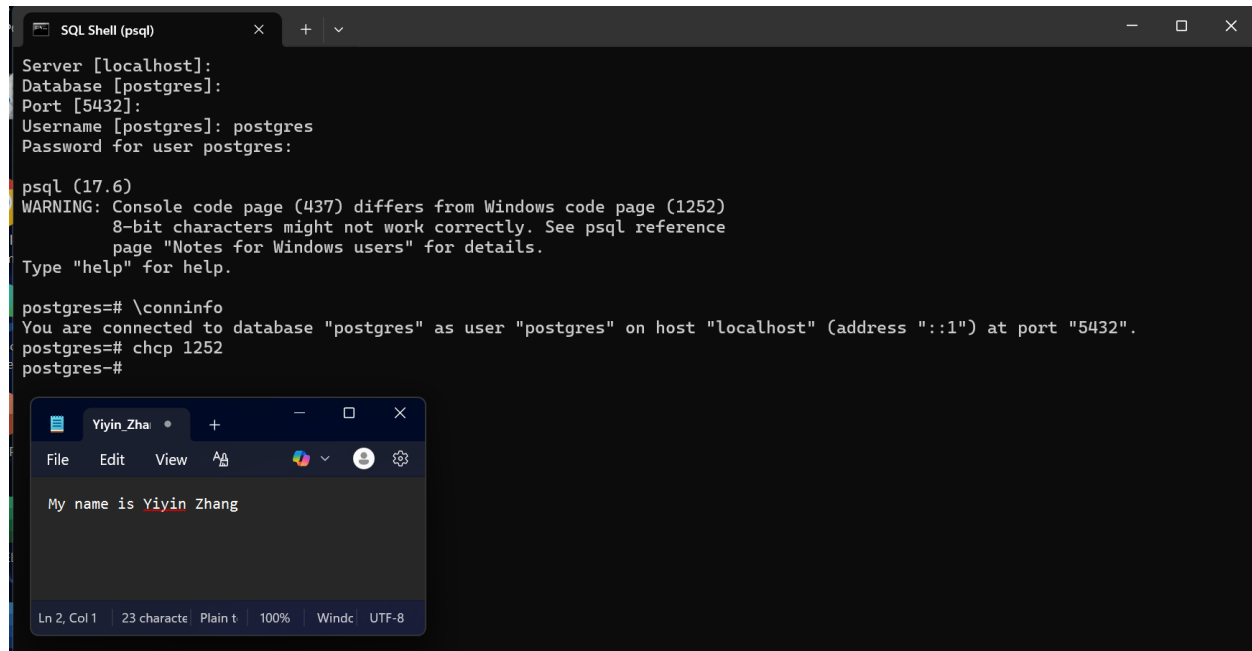
```
[1]: print("Hello from Yiyin Zhang")  
Hello from Yiyin Zhang
```

A terminal window is open in the foreground, showing a traceback for a 403 Forbidden error. The traceback indicates the error occurred in the file "C:\Users\yiyin\anaconda3\Lib\site-packages\tornado\web.py", line 1848, in _execute. The error message is "HTTP 403: missing nucleus_token".

A code editor window is also open, showing the following Python code:

```
My name is Yiyin Zhang
```

PostgreSQL



The image shows a PostgreSQL SQL Shell window and a text editor window. The SQL Shell window is titled "SQL Shell (psql)" and shows the following text:

```
Server [localhost]:
Database [postgres]:
Port [5432]:
Username [postgres]: postgres
Password for user postgres:

psql (17.6)
WARNING: Console code page (437) differs from Windows code page (1252)
8-bit characters might not work correctly. See psql reference
page "Notes for Windows users" for details.
Type "help" for help.

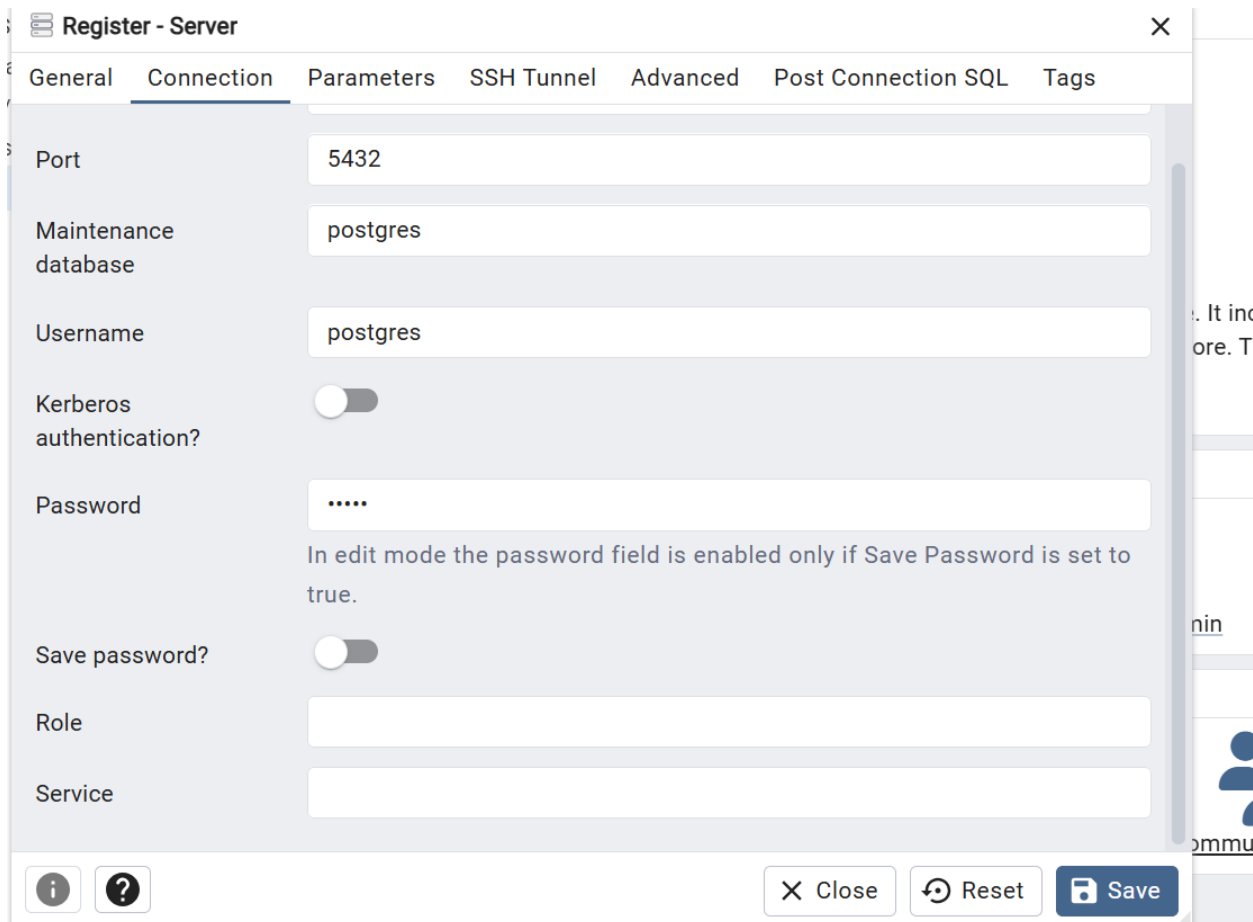
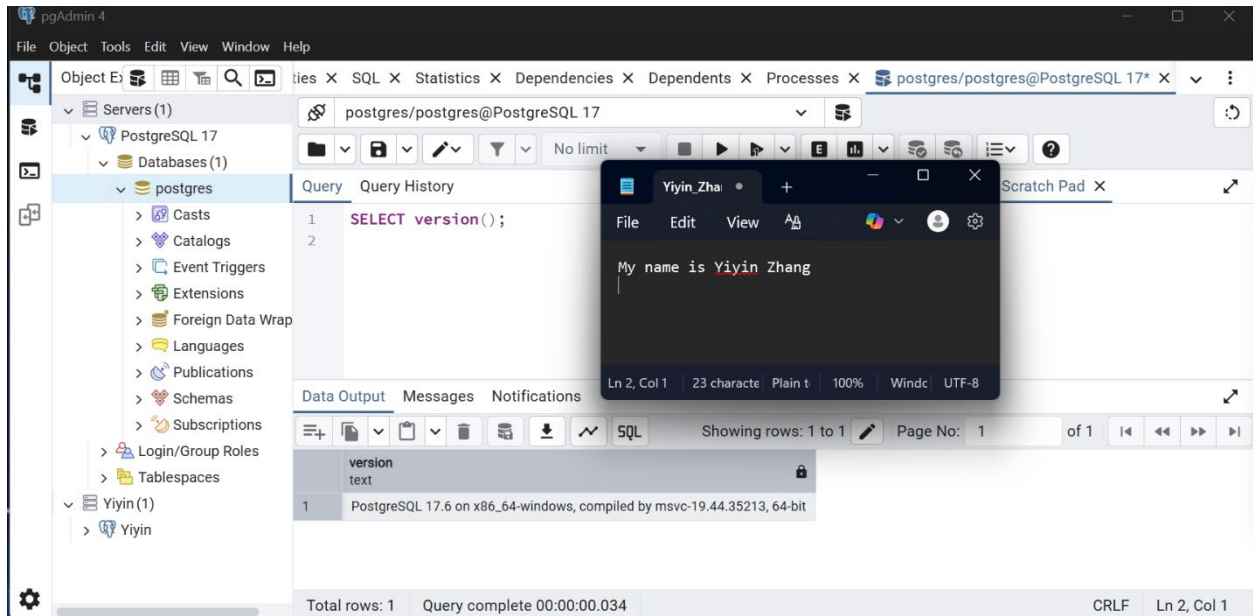
postgres=# \conninfo
You are connected to database "postgres" as user "postgres" on host "localhost" (address "::1") at port "5432".
postgres=# chcp 1252
postgres=#
```

The text editor window is titled "Yiyin_Zha" and shows the following text:

```
My name is Yiyin Zhang
```

The text editor window also shows a status bar at the bottom: "Ln 2, Col 1 | 23 character | Plain t | 100% | Windc | UTF-8".

PgAdmin4



Apache Spark

The image is a composite of three screenshots illustrating the setup and execution of Apache Spark in a Jupyter notebook environment.

Top Left: Terminal Window
The terminal shows the installation of Spark and JupyterLab extensions. The command `conda activate spark11` is used to activate the environment. Then, `pip install findspark` is run. The output shows that `findspark` version 2.0.1 is installed. Finally, `jupyter notebook` is run, and the output shows that various extensions (jupyter_lsp, jupyter_server_terminals, jupyterlab, notebook, notebook_shim, jupyterlab_widgets) are successfully linked and loaded.

Top Right: Jupyter Notebook Interface
The notebook shows the following code in a cell:

```
.master("local[*]") \
.getOrCreate()

print("Spark version =", spark.version)

df = spark.range(5)
df.show()
```

The output of the cell is:

```
Spark version = 3.5.6
+----+
| id |
+----+
|  0 |
|  1 |
|  2 |
|  3 |
|  4 |
+----+
```

Bottom: Web Browser View
The browser shows the JupyterLab interface at `localhost:8888/notebooks/Untitled1.ipynb?`. The left sidebar shows the Anaconda Toolbox with options like "Create a New Project", "Create a New Notebook", "My Projects", "Code Snippets", and "Anaconda AI Assistant". The main area shows the same code and output as the Jupyter Notebook interface above.

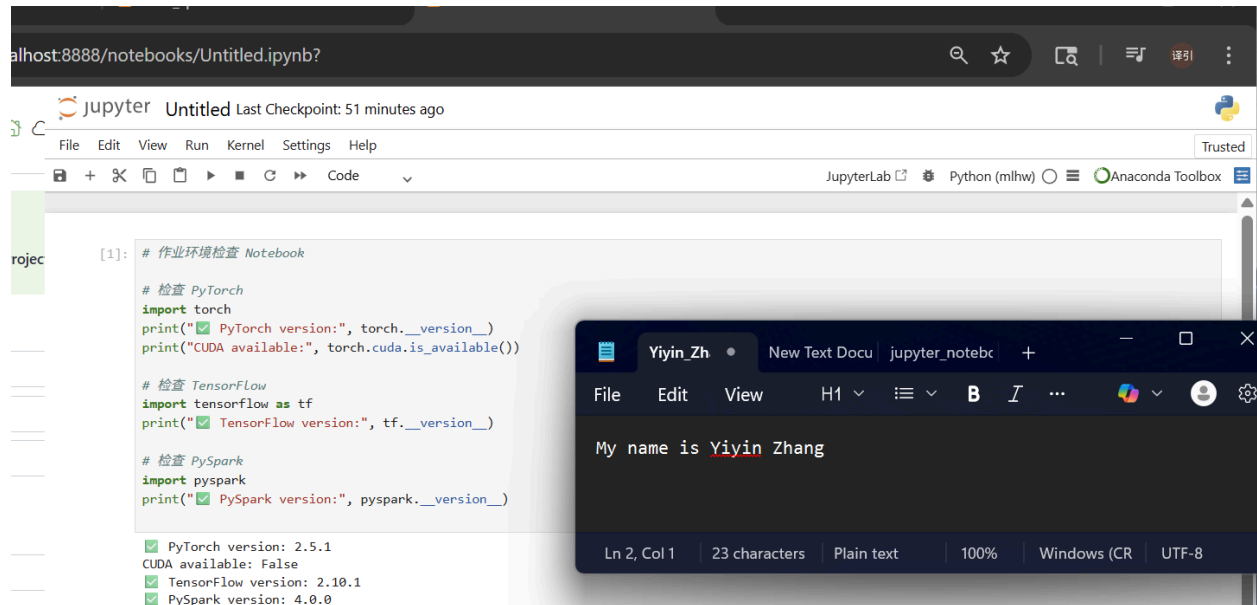
Tensorflow

The screenshot shows a Jupyter Notebook interface in a web browser at `localhost:8888/notebooks/test_tensorflow.ipynb`. The notebook has a menu bar with File, Edit, View, Run, Kernel, Settings, and Help. Below the menu is a toolbar with icons for saving, opening, and running code. The notebook content shows two code cells. The first cell imports TensorFlow and prints its version, which is 2.10.0. The second cell imports TensorFlow, defines two 2x2 matrices `a` and `b`, and prints the result of their matrix multiplication. The output of the second cell is a matrix multiplication result: `[[19 22] [43 50]]`.

Below the notebook, there is a text editor window titled "Yiyin_Zh" with the text "My name is Yiyin Zhang".

On the right side, the terminal output of the Jupyter Notebook is visible. It shows the kernel starting, connecting to the kernel, and the execution of the code cells. The output of the first cell is `2.10.0`. The output of the second cell is `Matrix multiplication result:\n [[19 22] [43 50]]`. The terminal also shows several warnings about missing GPU libraries, such as `Could not load dynamic library 'cudart64_110.dll'` and `Could not load dynamic library 'cublas64_11.dll'`.

Pytorch



The screenshot displays a JupyterLab environment. The main notebook, titled "Untitled", contains a code cell with the following Python code for environment verification:

```
[1]: # 作业环境检查 Notebook

# 检查 PyTorch
import torch
print("✅ PyTorch version:", torch.__version__)
print("CUDA available:", torch.cuda.is_available())

# 检查 TensorFlow
import tensorflow as tf
print("✅ TensorFlow version:", tf.__version__)

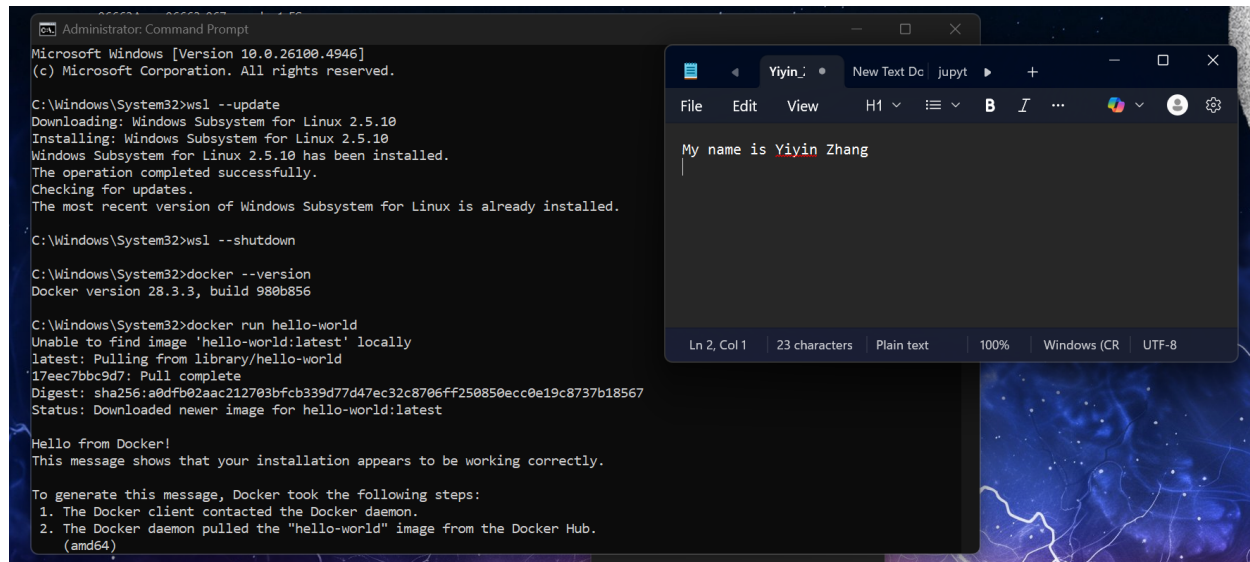
# 检查 PySpark
import pyspark
print("✅ PySpark version:", pyspark.__version__)
```

The output of the code cell shows the following results:

- ✅ PyTorch version: 2.5.1
- CUDA available: False
- ✅ TensorFlow version: 2.10.1
- ✅ PySpark version: 4.0.0

Overlaid on the notebook is a text editor window titled "Yiyin_Zh" with the text "My name is Yiyin Zhang". The text editor shows "Ln 2, Col 1" and "23 characters".

Docker



The screenshot shows a Windows desktop with a Command Prompt window and a text editor window. The Command Prompt window is titled "Administrator: Command Prompt" and displays the following output:

```
Microsoft Windows [Version 10.0.26100.4946]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>wsl --update
Downloading: Windows Subsystem for Linux 2.5.10
Installing: Windows Subsystem for Linux 2.5.10
Windows Subsystem for Linux 2.5.10 has been installed.
The operation completed successfully.
Checking for updates.
The most recent version of Windows Subsystem for Linux is already installed.

C:\Windows\System32>wsl --shutdown

C:\Windows\System32>docker --version
Docker version 28.3.3, build 980b856

C:\Windows\System32>docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
Digest: sha256:a0dfb02aac212703bfc339d77d47ec32c8706ff250850ecc0e19c8737b18567
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
```

The text editor window is titled "Yiyin" and contains the text "My name is Yiyin Zhang". The editor's status bar shows "Ln 2, Col 1", "23 characters", "Plain text", "100%", "Windows (CR)", and "UTF-8".

Part-II

Get familiar with GitHub

README.mdUpdate X in STEP and README.md2 minutes agoNo packages published
[Publish your first package](#)

READMEMIT license

Introduction to GitHub

Get started using GitHub in less than an hour.

Finish

Congratulations, you've completed this course and joined the world of developers!

Here's a recap of your accomplishments:

- You learned about GitHub, repositories, branches, commits, and pull requests.
- You created a branch, a commit, and a pull request.
- You merged a pull request.
- You made your first contribution! 🎉

What's next?

If you'd like to make a profile README, use the quickstart instructions below or follow the instructions in the [Managing your profile README](#) article.

1. Make a new public repository with a name that matches your GitHub username.
2. Create a file named `README.md` in its root. The "root" means not inside any folder in your repository.
3. Edit the contents of the `README.md` file.
4. If you created a new branch for your file, open and merge a pull request on your branch.
5. Lastly, we'd love to hear what you thought of this course [in our discussion board](#).

Check out these resources to learn more or get involved:

- Are you a student? Check out the [Student Developer Pack](#).
- [Take another GitHub Skills course](#).
- [Read the GitHub Getting Started docs](#).
- To find projects to contribute to, check out [GitHub Explore](#).

Get help: [Post in our discussion board](#) • [Review the GitHub status page](#)

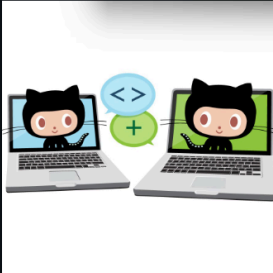
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Yiyin_ZhangNew Text Documentjupyter+

FileEditViewH1≡BIL...100%Windows (CR)UTF-8

My name is Yiyin Zhang
Andrew_ID: yiyinz

Ln 2, Col 1841 charactersPlain text100%Windows (CR)UTF-8



github.com/Elyanny/skills-communicate-using-markdown

index.md Update index.md 3 minutes ago

README MIT license

Communicate using Markdown

Organize ideas and collaborate using Markdown, a lightweight language for text formatting.

Finish

Congratulations friend, you've completed this course!

Here's a recap of all the tasks you've accomplished in your repository:

1. You learned about Markdown, headings, images, code examples, and task lists.
2. You created and merged a Markdown file.
3. You learned an essential GitHub skill. 🎉



What's next?

- You can enable GitHub Pages and see your Markdown file as a website!
 - i. Under your repository name at the upper right, click .
 - ii. Then on the lower left, click **Pages** in the **Code and automation** section.
 - iii. In the **GitHub Pages** section, ensure "Deploy from a branch" is selected from the **Source** drop-down menu, and then select `main` from the **Branch** drop-down menu as your GitHub Pages publishing source.
 - iv. Click the **Save** button.
 - v. Wait about 30 seconds then refresh the page. When you see "Your site is published at ..." you can click on the link to see your published site.
- Learn more about [Markdown](#).
- We'd love to hear what you thought of this course [in our discussion board](#)
- [Take another GitHub Skills course](#).
- [Read the GitHub Getting Started docs](#).
- To find projects to contribute to, check out [GitHub Explore](#).

Get help: [Post in our discussion board](#) • [Review the GitHub status page](#)

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Packages

No packages published
[Publish your first package](#)

Contributors 2

- github-actions[bot]
- Elyanny

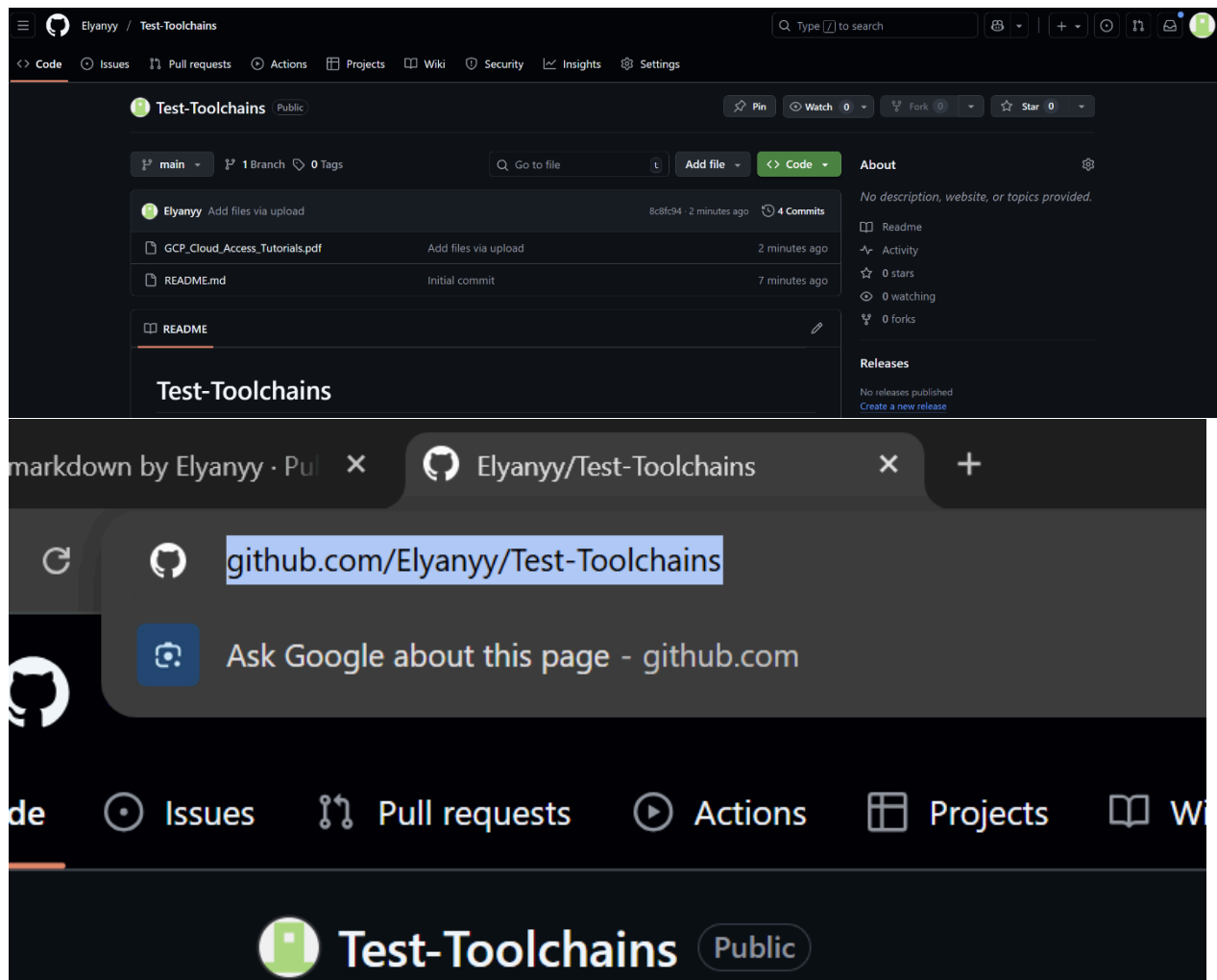
Add new tab

Yiyin J. New Text Do jupyter

File Edit View H1 H2 H3 B I ... 100% Windows (CR) UTF-8

```
My name is Yiyin Zhang
Andrew_ID: yiyinz
```

Following task



My URL: <https://github.com/Elyanyy/Test-Toolchains>